

Qatar-1b

R-2716

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_120619 · created 2026-09-01T01:45:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456097.78094 +/- 0.0002 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.191 +/- 0.056
Transit depth (Rp/Rs)^2	3.66 +/- 2.16 [%]
Orbital Inclination (inc)	89.0 +/- 2.82
Airmass coefficient 1 (a1)	1.32 +/- 0.17
Airmass coefficient 2 (a2)	-0.23 +/- 0.17
Scatter in the residuals of the lightcurve fit is	13.61 %
Best Comparison Star	#1 - [327, 175]
Optimal Aperture	5.23
Optimal Annulus	22.38
Transit Duration (day)	0.0832 +/- 0.0065

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.191 ± 0.056 vs 0.1463 (deviation 0.66σ)
Duration fit vs published	0.0832 d vs 0.0692 d (deviation 1.79σ)
Mid-time O–C	0.2 min
Depth SNR	2.8
Residual scatter	13.61 %

Light curve

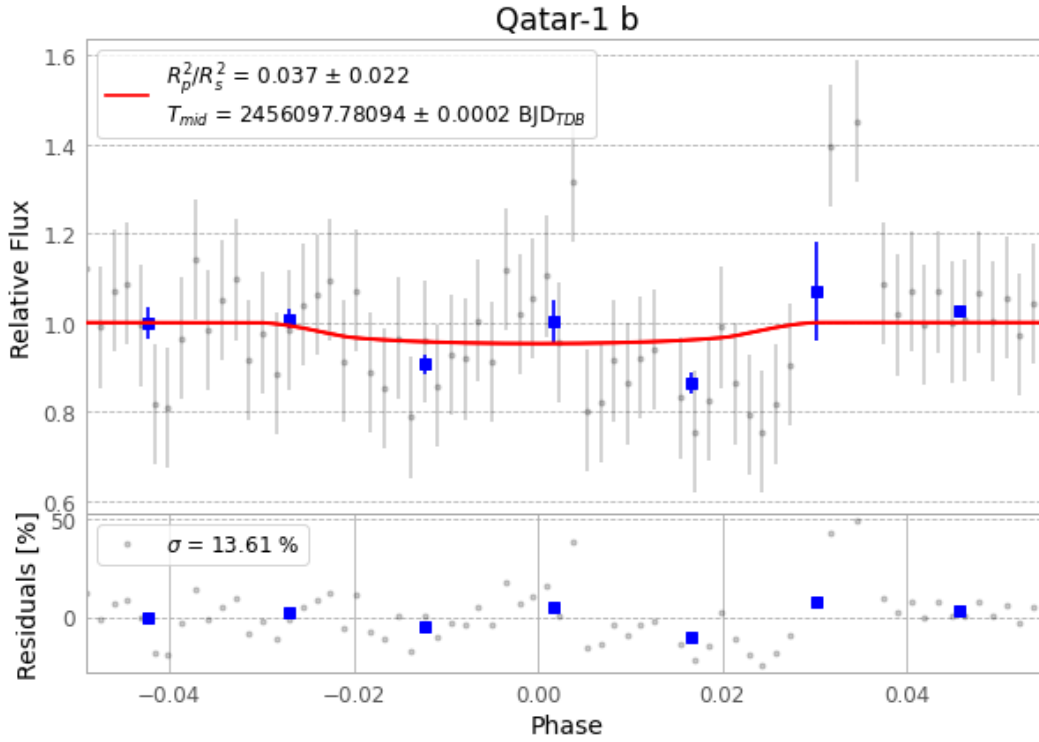


Figure 1 — FinalLightCurve_Qatar-1 b_2012-06-18.png (SHA-256 c8c2197cf931e98c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [218, 380], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5c364a7cc6bf01bc...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), Qatar-1120619050312.FITS → Qatar-1120619083614.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-120619.log (71d3150a0e12...), FinalLightCurve_Qatar-1 b_2012-06-18.png (c8c2197cf931...), FinalLightCurve_Qatar-1 b_2012-06-18.csv (b5f0608f2587...)

Compute resources

Wall time 133.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-01 01:45Z.